Card Fraud Triage with Gaussian Mixtures and Deep Autoencoders

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Abstract—Fraud triage requires rankings that surface rare frauds under limited review capacity. We compare a full-covariance Gaussian mixture model (GMM), a 12-dimensional autoencoder (AE), and a validation-selected blend on 56,962 chronologically held-out transactions containing 75 frauds. The base models are fitted on legitimate transactions. The AE uses deduplicated training features. The GMM achieves average precision of 0.4833 versus 0.2453 for the AE, while the AE leads in ROC-AUC (0.9274 versus 0.9106). Validation average precision selects a GMM weight of 1.00, so the blend reproduces the GMM ranking. At a validation false-positive budget of 0.10%, the GMM detects 57 frauds with 35 false alerts, and the AE detects 22 with 30 false alerts. Every AE hit is also a GMM hit at that cutoff and in the first 100 reviews.

Index Terms—card fraud, anomaly detection, Gaussian mixture models, autoencoders, alert triage, average precision

I. Introduction and Operational Constraints

The European card transaction benchmark contains 492 frauds among 284,807 transactions [1]. Analysts can inspect only a fraction of alerts. A ranking can discriminate well over the full score range while returning too many false alerts near its head. We therefore report average precision (AP), ROC-AUC, recall and precision under fixed false-positive budgets, and precision and recall under fixed queue sizes. The held-out window contains 75 frauds, so one recovered fraud changes recall by about 1.33 percentage points.

II. Related Work

Dal Pozzolo et al. [1] use an imbalanced card dataset to study how undersampling changes the class prior and biases predicted probabilities. A prior correction can change calibration while preserving the ordering. We preserve the observed class proportions in validation and test and interpret detector outputs as anomaly scores. Dempster et al. [2] describe expectation-maximization (EM) for latent assignments, including finite mixtures. Sakurada and Yairi [3] compare nonlinear autoencoder reconstruction error against linear and kernel PCA on simulated dynamics and spacecraft telemetry. We apply reconstruction error to card transaction features and compare it with mixture likelihood under a review budget.

III. Anti-Leakage Data Pipeline and Duplicate Audit

A stable sort on elapsed Time precedes a chronological 60/20/20 split. Training uses only legitimate rows. We drop Time after splitting because elapsed time would expose a non-overlapping range at deployment. The 29 scoring features are the supplied PCA components V1–V28 and Amount. The Phase 1 scaler and GMM use the original 170,524 legitimate training rows. Phase 2 removes 5,245 repeated 29-feature vectors (3.076%), leaving 165,279 rows for the AE and its scaler. Repeated feature vectors can overweight recurring patterns in the training loss. Validation (56,961 rows, 57 frauds) and test retain all arrivals.

In the 8-dimensional ablation, legitimate validation MSE decreased from 0.5953 to 0.5929 (Table I). Each training partition has its own standardization, so the observed change reflects both altered sample weights and scaling. It is a single-run comparison, and validation AP in this ablation fell from 0.0745 to 0.0683.

TABLE I
Duplicate ablation (8D AE). The 99% cutoff is derived from the respective legitimate training scores.

Train set	Train MSE	Val MSE	Cutoff	Val AP	Val ROC
Full	.3471	.5953	1.7199	.0745	.9445
Dedup.	.3506	.5929	1.7452	.0683	.9449

IV. Comparative Methodology

A. Full-covariance Gaussian mixture

For standardized input $x \in \mathbb{R}^D$, $D = 29$, the GMM is

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x; \mu_k, \Sigma_k), \quad K = 10, \tag{1}$$

$$\mathcal{N}(x; \mu, \Sigma) = \frac{\exp[-(x-\mu)^\top \Sigma^{-1}(x-\mu)/2]}{(2\pi)^{D/2}|\Sigma|^{1/2}}, \tag{2}$$

where $\pi_k \geq 0$ and $\sum_k \pi_k = 1$. EM estimates latent responsibilities with the current parameters,

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)}. \tag{3}$$

For N training rows, the update with covariance regularization is

$$N_k = \sum_i r_{ik}, \qquad \pi_k^+ = N_k/N, \tag{4}$$

$$\mu_k^+ = N_k^{-1} \sum_i r_{ik} x_i, \tag{5}$$

$$\Sigma_k^+ = N_k^{-1} \sum_i r_{ik}(x_i - \mu_k^+)(x_i - \mu_k^+)^\top + \lambda I. \tag{6}$$

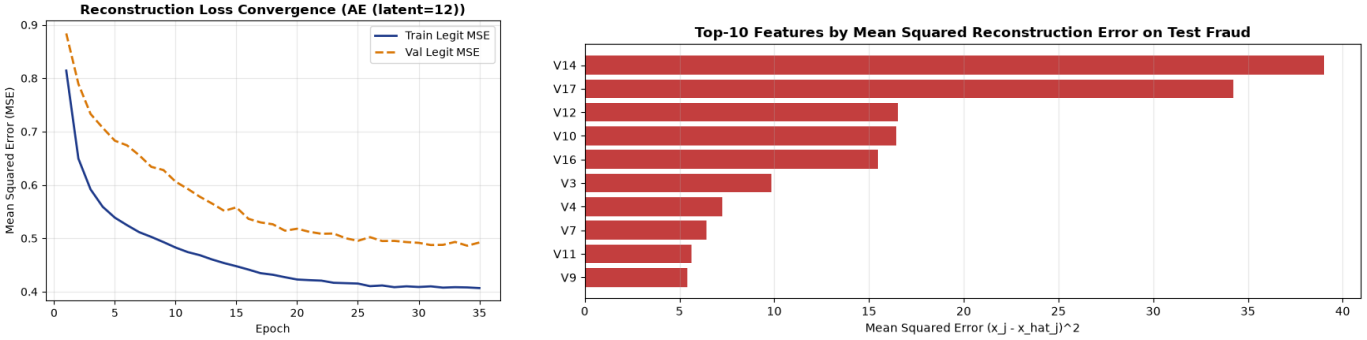


Fig. 1. Saved training-notebook outputs. Left, standard 12D AE loss by epoch. Right, the ten features with highest mean squared reconstruction error on test frauds.

The saved fit uses five initializations and $\lambda = 10^{-6}$. Its score is $s_G(x) = -\log p(x)$.

B. Nonlinear autoencoder

The standard AE uses widths $29 \to 22 \to 14 \to 12 \to 14 \to 22 \to 29$. Hidden blocks use batch normalization and LeakyReLU (slope 0.1), with dropout of 0.10 after the first hidden block of each half. The bottleneck and reconstruction layers have linear outputs. For encoder f_θ and decoder g_ϕ, training minimizes

$$\mathcal{L}(\theta, \phi) = \frac{1}{ND} \sum_{i=1}^{N} \|x_i - g_\phi(f_\theta(\tilde{x}_i))\|_2^2. \tag{7}$$

For standard training, $\tilde{x}_i = x_i$. Denoising runs add independent Gaussian noise with standard deviation 0.05 in standardized input units. At inference, the clean standardized input gives the anomaly score $s_A(x) = D^{-1}\|x - g_\phi(f_\theta(x))\|_2^2$. We use the saved 12-dimensional standard AE, which led the sweep on validation AP. The sweep trained with AdamW and early stopping on legitimate validation MSE.

C. Score fusion and triage

We fit means μ_m and population standard deviations σ_m on all validation scores for each detector m. The blend is

$$s_H(x) = \alpha \frac{s_G(x) - \mu_G}{\sigma_G} + (1 - \alpha) \frac{s_A(x) - \mu_A}{\sigma_A}. \tag{8}$$

We select α by validation AP from $\{0, 0.25, 0.50, 0.75, 1\}$. The base fits use legitimate training data. Validation labels enter blend selection and identify legitimate validation rows for FPR calibration.

For target FPR q, sort the $N_0 = 56{,}904$ legitimate validation scores in descending order and flag scores strictly greater than the first excluded score, at rank $\lfloor qN_0 \rfloor + 1$. This limits validation false positives to $\lfloor qN_0 \rfloor$ even with tied scores. We freeze the cutoff before scoring test. For fixed queues, we rank the test stream by descending score and use CSV order to break ties.

V. Unsupervised Manifold Evaluation and Triage Results

Table II compares bottlenecks and denoising variants. The 12-dimensional standard AE has the lowest legitimate validation MSE (0.4860) and highest validation AP (0.1222). The 16-dimensional model reconstructs the training sample more closely but gives higher validation MSE. Fig. 1 shows the original convergence and attribution plots. V14 and V17 exhibit fraud-to-legitimate mean squared reconstruction error ratios of 123.28 and 137.34 (Table III). These PCA coordinates localize reconstruction error.

TABLE II
Bottleneck sweep. MSE uses legitimate rows. The 99% cutoff uses the training split.

Model	d	Noise	Params	Train MSE	Val MSE	Cutoff	Val AP	Val ROC
AE	4	0.00	2,253	0.4907	0.7418	2.3849	0.0478	0.9357
AE	8	0.00	2,369	0.3506	0.5929	1.7452	0.0683	0.9449
AE	12	0.00	2,485	0.2967	0.4860	1.4322	0.1222	0.9467
AE	16	0.00	2,601	0.2884	0.5121	1.5165	0.0785	0.9415
DAE	8	0.05	2,369	0.3654	0.6070	1.8661	0.0616	0.9369
DAE	12	0.05	2,485	0.3068	0.5405	1.4879	0.0846	0.9531

TABLE III
Top ten features by mean squared error on test frauds.

Feature	Legit MSE	Fraud MSE	Fraud/legit
V14	0.3164	38.9999	123.28
V17	0.2494	34.2506	137.34
V12	0.2763	16.5218	59.79
V10	0.2267	16.4190	72.42
V16	0.3648	15.4536	42.36
V3	1.2007	9.8601	8.21
V4	0.3461	7.2553	20.97
V7	0.3577	6.4272	17.97
V11	0.6946	5.6463	8.13
V9	0.3259	5.4359	16.68

The geometry audit compares hard GMM assignments in standardized 29-dimensional space with 10-cluster k-means assignments in the 12-dimensional latent space. The metrics use a seeded sample of 10,000 test rows and no fraud labels. The latent representation gives higher

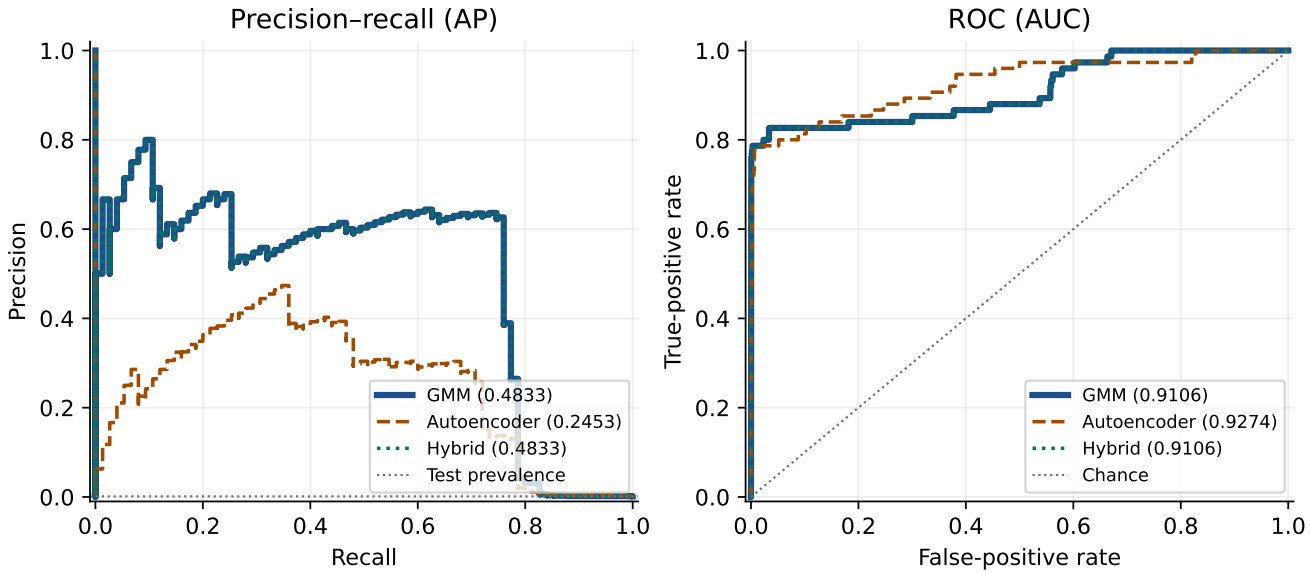


Fig. 2. Held-out precision–recall and ROC curves. The hybrid overlaps the GMM because the validation sweep selects $\alpha = 1$. The PR reference is test fraud prevalence (75/56,962).

Silhouette and Calinski–Harabasz and lower Davies–Bouldin values (Table IV). The clustering algorithm also changes, so the comparison does not isolate the effect of representation. The pseudo-label agreement experiment (Tables V and VI) applies training score quantiles to test. It is a distinct protocol from the validation false-positive budgets.

TABLE IV
Intrinsic cluster metrics (10 clusters per space).

Space / assignments	Silhouette ↑	DB ↓	CH ↑
29D / GMM	0.0364	4.4603	168.82
12D / k-means	0.1836	1.5472	1048.19

TABLE V
Training-quantile pseudo-label agreement. Cutoffs come from scores on deduplicated legitimate training rows. Alert counts and agreement use test.

Quantile	GMM cut	AE cut	GMM	AE	Both	κ	Jaccard	Spear. ρ
99.0%	52.056	1.4322	880	854	495	0.5643	0.3995	0.5049
99.5%	57.504	2.0401	530	397	283	0.6074	0.4394	0.5049
99.9%	86.735	5.8213	120	88	72	0.6918	0.5294	0.5049

TABLE VI
Test audit of 99.9% train-quantile pseudo-labels.

Rule	Alerts	TP	FP	Prec. (%)	Recall (%)
GMM	120	57	63	47.50	76.00
AE	88	34	54	38.64	45.33
Both	72	34	38	47.22	45.33

Table VII shows the validation sweep. At $\alpha = 1$, the hybrid is the GMM score under positive affine rescaling, which leaves rankings and corresponding alert decisions unchanged. On test (Table VIII and Fig. 2), GMM leads on AP and AE leads on ROC-AUC. Phase 1 reported trapezoidal PR area of 0.4736 for GMM. Here we use scikit-learn's non-interpolated AP, which weights precision by increments in recall.

TABLE VII
Blend selection on validation.

GMM weight α	AP	ROC-AUC
0.00	0.1222	0.9467
0.25	0.1550	0.9476
0.50	0.1842	0.9458
0.75	0.2437	0.9402
1.00	0.4782	0.9351

TABLE VIII
Global ranking on the held-out test split.

Model	AP	ROC-AUC
GMM	0.4833	0.9106
AE	0.2453	0.9274
Hybrid	0.4833	0.9106

Under the tightest validation FPR budget, GMM reaches 76.0% recall and 61.96% precision. AE reaches 29.33% recall and 42.31% precision. At the 0.50% budget, the AE has higher precision and lower recall than GMM. Their realized test FPRs differ, so Table IX includes both the target and realized rate. In equal-size queues, GMM

leads at $K = 50, 100, 200$ and ties the AE at $K = 500$, where each retrieves 59 frauds (Table X).

TABLE IX
Validation-only cutoffs and realized test triage. Each model flags 56, 284 and 569 legitimate validation rows at the respective budgets (0.0984%, 0.4991%, 0.9999%).

Model	Val (%)	Cutoff	TP	FP	FPR (%)	Recall (%)	Prec. (%)
GMM	0.10	114.749583	57	35	0.0615	76.00	61.96
GMM	0.50	68.723566	59	192	0.3375	78.67	23.51
GMM	1.00	60.692267	59	346	0.6082	78.67	14.57
AE	0.10	9.780338	22	30	0.0527	29.33	42.31
AE	0.50	3.057487	53	141	0.2479	70.67	27.32
AE	1.00	2.016608	55	346	0.6082	73.33	13.72
Hybrid	0.10	0.983374	57	35	0.0615	76.00	61.96
Hybrid	0.50	0.665640	59	192	0.3375	78.67	23.51
Hybrid	1.00	0.610197	59	346	0.6082	78.67	14.57

TABLE X
Fixed review queues. Cells show precision / recall in percent.

K	GMM	AE	Hybrid
50	58.0 / 38.7	40.0 / 26.7	58.0 / 38.7
100	57.0 / 76.0	35.0 / 46.7	57.0 / 76.0
200	29.0 / 77.3	26.5 / 70.7	29.0 / 77.3
500	11.8 / 78.7	11.8 / 78.7	11.8 / 78.7

VI. Error Analysis and Conclusion

At the 0.10% validation budget, the models catch 22 frauds in common and GMM catches 35 additional frauds. In Top 100 the counts are 35 and 22. Both rules miss the same 18 cases and neither produces an AE-only hit (Table XI). The GMM-only cases have more negative median V14, V17 and V12 than shared misses at both operating rules (Table XII). Median Amount for GMM-only hits is 10.70 at the cutoff and 7.13 in the queue, versus 7.80 for shared misses, with overlapping interquartile ranges. The executed notebook lists the complete row indices and interquartile ranges for each group.

TABLE XI
Fraud overlap by operating rule. Each row totals 75.

Rule	Both	GMM only	AE only	Missed
Val FPR 0.10%	22	35	0	18
Top 100	35	22	0	18

TABLE XII
Median raw features by fraud detection group. An em dash denotes an empty group.

Rule	Group	n	Amount	V14	V17	V12
Val FPR 0.10%	Both	22	1.00	-11.93	-5.73	-6.40
Val FPR 0.10%	GMM only	35	10.70	-6.28	-2.52	-3.81
Val FPR 0.10%	AE only	0	—	—	—	—
Val FPR 0.10%	Missed by both	18	7.79	-1.89	1.33	-0.17
Top 100	Both	35	1.00	-9.15	-5.13	-6.12
Top 100	GMM only	22	7.13	-5.94	-1.80	-3.12
Top 100	AE only	0	—	—	—	—
Top 100	Missed by both	18	7.79	-1.89	1.33	-0.17

On this time window, GMM yielded the strongest short-queue ranking and the validation-selected blend added no AE weight. The AE representation showed stronger intrinsic cluster geometry and a higher ROC-AUC. The contrast illustrates why ranking and review-budget metrics are needed alongside geometry. These findings come from one benchmark, one saved fit per method, and 75 held-out frauds. Calibration limits false alerts in validation, and realized false-positive rates can shift in future time windows.

References

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